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SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 08

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23rd July, 2026
Title of Assignment:	Dockerfile & Image Build
Theory:	Docker is a containerization platform that enables applications and their dependencies to be packaged into lightweight, portable containers. A Dockerfile contains the instructions required to build a Docker image, while a Docker image serves as a template for creating containers. A Docker container is a running instance of an image that provides an isolated environment for executing applications consistently across different systems. In this experiment, a simple Flask web application was containerized by creating the required application files, building a Docker image, running the container, and verifying the application through a web browser. **Commands Used** • mkdir my-flask-app – Created a new directory for the Flask application. • cd my-flask-app – Navigated to the project directory. • docker build -t my-flask-app . – Built a Docker image named my-flask-app from the Dockerfile. • docker build --no-cache -t my-flask-app . – Rebuilt the Docker image without using cached layers after updating the dependencies. • docker run -p 5000:5000 my-flask-app – Started the Docker container and mapped port 5000 of the container to port 5000 on the host machine for browser access.

Implementation:

Step 1: Create the Flask Application

Created Flask Application — Created the my-flask-app project directory and added the required application files. Implemented the app.py file containing a simple Flask web application that displays the message "Hello, Docker!" when accessed.

Step 2: Create the Requirements File

Created Requirements File — Created the requirements.txt file specifying the Flask dependency required to install the necessary packages for the Flask application.

Step 3 – Create the Dockerfile

Created Dockerfile — Created the Dockerfile specifying the Python base image, working directory, application dependencies, exposed port, and startup command required to containerize the Flask application.

Step 4 – Build the Docker Image

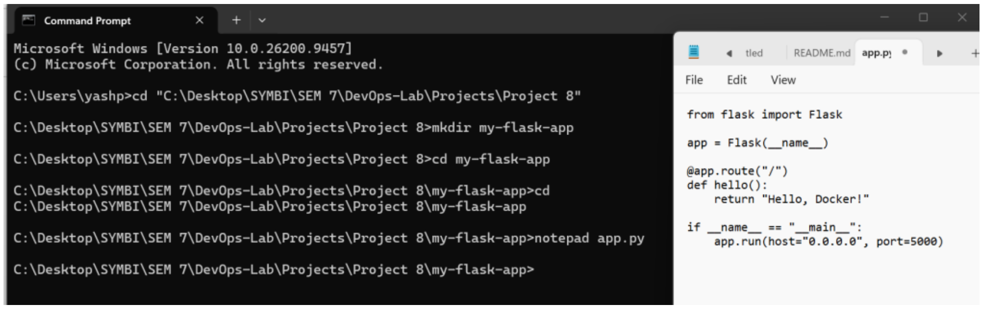
Built Docker Image — Successfully built the Docker image named my-flask-app using the Dockerfile. Docker completed all build stages and created the container image successfully.

Step 5 – Run the Docker Container

Executed Docker Container — Successfully started the Docker container and launched the Flask application. The application is running on port 5000 and is accessible through the host machine.

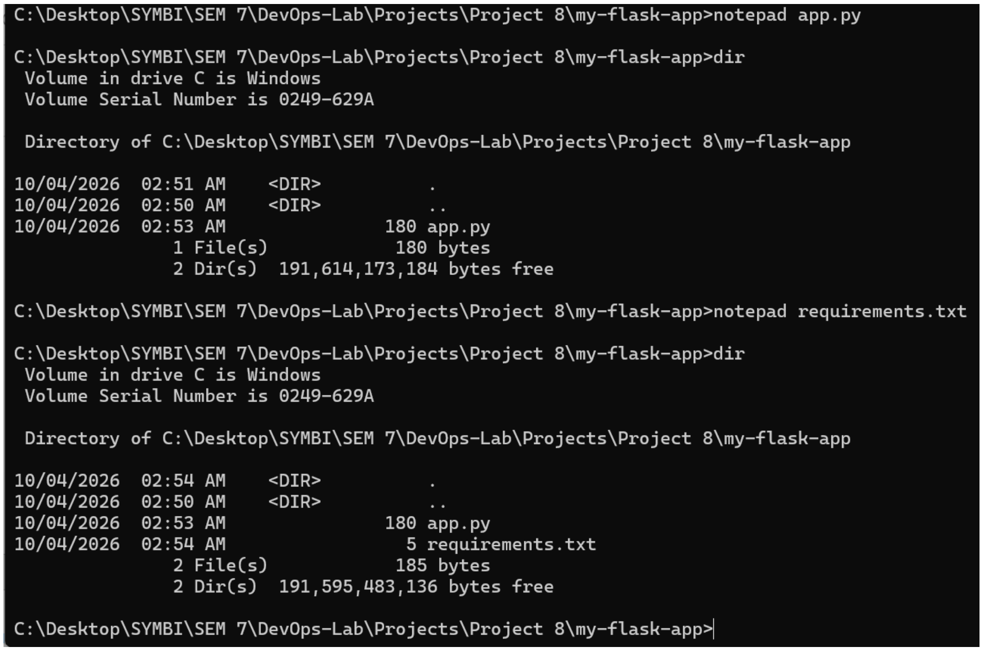
Step 6: Verify the Flask Application

Verified Dockerized Flask Application — Opened the Flask application in a web browser using http://localhost:5000 and successfully verified that the Docker container served the "Hello, Docker!" webpage.

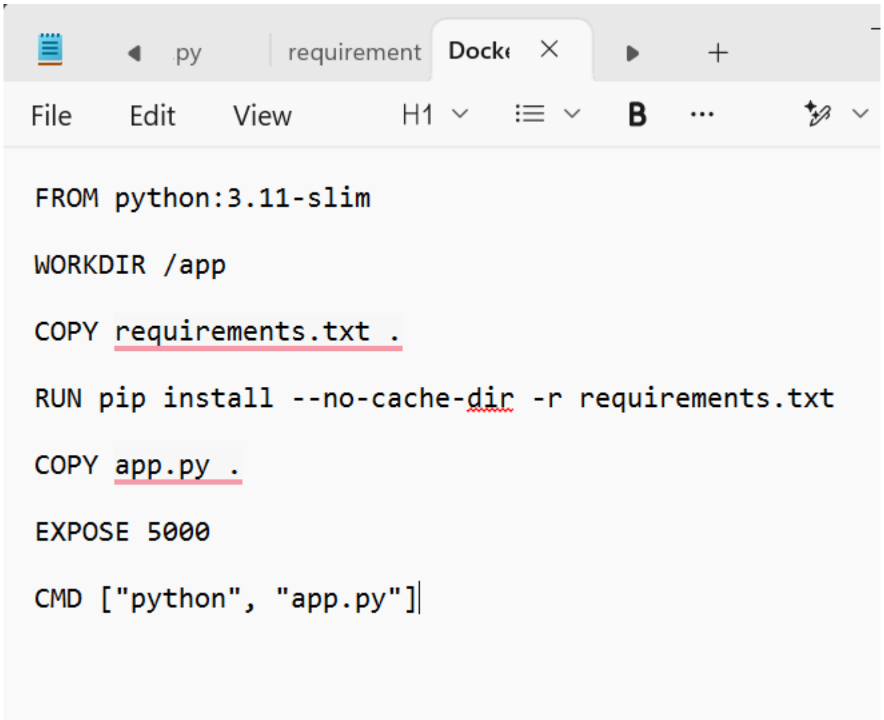


```
Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\yashp>cd "C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8"

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8>mkdir my-flask-app

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8>cd my-flask-app

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>cd
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>notepad app.py

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>
```

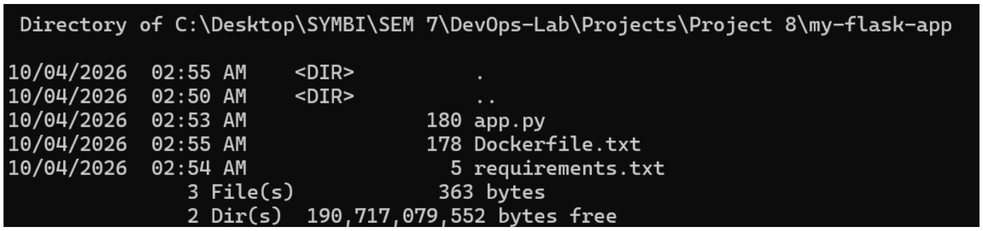
```python
from flask import Flask

app = Flask(__name__)

@app.route("/")
def hello():
    return "Hello, Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```



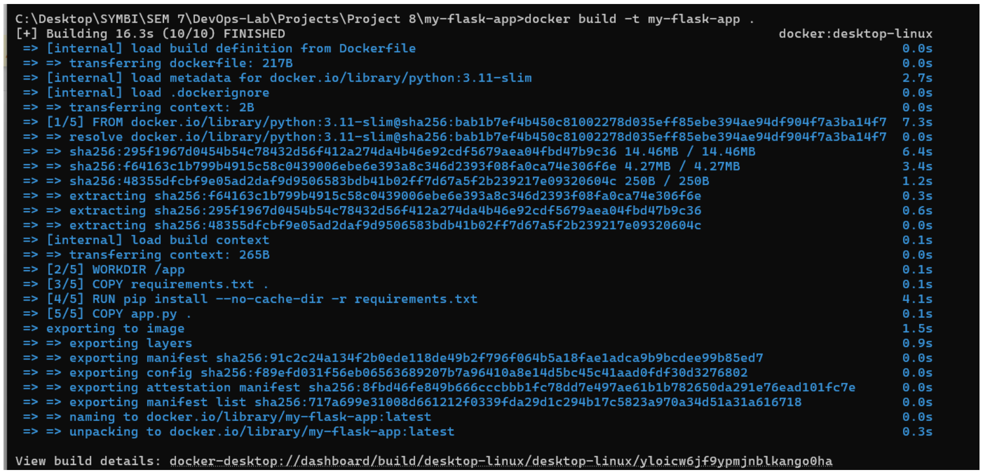
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>notepad app.py

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>dir
 Volume in drive C is Windows
 Volume Serial Number is 0249-629A

 Directory of C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app

10/04/2026  02:51 AM    <DIR>          .
10/04/2026  02:50 AM    <DIR>          ..
10/04/2026  02:53 AM               180 app.py
               1 File(s)            180 bytes
               2 Dir(s)  191,614,173,184 bytes free

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>notepad requirements.txt

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>dir
 Volume in drive C is Windows
 Volume Serial Number is 0249-629A

 Directory of C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app

10/04/2026  02:54 AM    <DIR>          .
10/04/2026  02:50 AM    <DIR>          ..
10/04/2026  02:53 AM               180 app.py
10/04/2026  02:54 AM                 5 requirements.txt
               2 File(s)            185 bytes
               2 Dir(s)  191,595,483,136 bytes free

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>
```



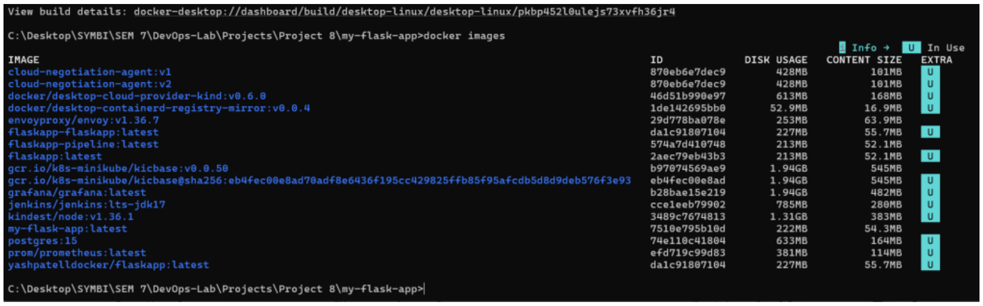
```
FROM python:3.11-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY app.py .

EXPOSE 5000

CMD ["python", "app.py"]
```



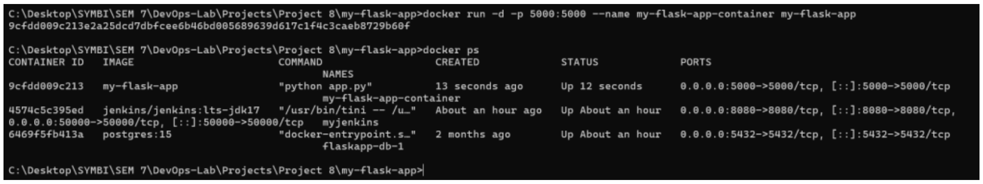
```
 Directory of C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app

10/04/2026  02:55 AM    <DIR>          .
10/04/2026  02:50 AM    <DIR>          ..
10/04/2026  02:53 AM               180 app.py
10/04/2026  02:55 AM               178 Dockerfile.txt
10/04/2026  02:54 AM                 5 requirements.txt
               3 File(s)            363 bytes
               2 Dir(s)  190,717,079,552 bytes free
```



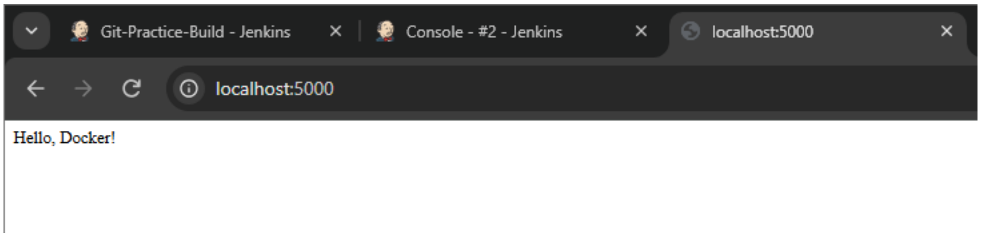
```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>docker build -t my-flask-app .
[+] Building 16.3s (10/10) FINISHED                                                                        docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                       0.0s
 => => transferring dockerfile: 217B                                                                                       0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                        2.7s
 => [internal] load .dockerignore                                                                                          0.0s
 => => transferring context: 2B                                                                                            0.0s
 => [1/5] FROM docker.io/library/python:3.11-slim@sha256:bab1b7ef4b450c81002278d035eff85ebe394ae94df904f7a3ba14f7  7.3s
 => => resolve docker.io/library/python:3.11-slim@sha256:bab1b7ef4b450c81002278d035eff85ebe394ae94df904f7a3ba14f7  0.0s
 => => sha256:295f1967d0454b54c78432d56f412a274da4b46e92cdf5679aea04fbd47b9c36 14.46MB / 14.46MB                           6.4s
 => => sha256:f64163c1b799b4915c58c0439006ebe6e393a8c346d2393f08fa0ca74e306f6e 4.27MB / 4.27MB                             3.4s
 => => sha256:48355dfcbf9e05ad2daf9d9506583bdb41b02ff7d67a5f2b239217e09320604c 250B / 250B                                 1.2s
 => => extracting sha256:f64163c1b799b4915c58c0439006ebe6e393a8c346d2393f08fa0ca74e306f6e                                  0.3s
 => => extracting sha256:295f1967d0454b54c78432d56f412a274da4b46e92cdf5679aea04fbd47b9c36                                  0.6s
 => => extracting sha256:48355dfcbf9e05ad2daf9d9506583bdb41b02ff7d67a5f2b239217e09320604c                                  0.0s
 => [internal] load build context                                                                                          0.1s
 => => transferring context: 265B                                                                                          0.0s
 => [2/5] WORKDIR /app                                                                                                     0.1s
 => [3/5] COPY requirements.txt .                                                                                          0.1s
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt                                                               4.1s
 => [5/5] COPY app.py .                                                                                                    0.1s
 => exporting to image                                                                                                     1.5s
 => => exporting layers                                                                                                    0.9s
 => => exporting manifest sha256:91c2c24a134f2b0ede118de49b2f796f064b5a18fae1adca9b9bcdee99b85ed7                          0.0s
 => => exporting config sha256:f89efd031f56eb06563689207b7a96410a8e14d5bc45c41aad0fdf30d3276802                            0.0s
 => => exporting attestation manifest sha256:8fbd46fe849b666cccbbb1fc78dd7e497ae61b1b782650da291e76ead101fc7e              0.0s
 => => exporting manifest list sha256:717a699e31008d661212f0339fda29d1c294b17c5823a970a34d51a31a616718                     0.0s
 => => naming to docker.io/library/my-flask-app:latest                                                                     0.0s
 => => unpacking to docker.io/library/my-flask-app:latest                                                                  0.3s

 View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/yloicw6jf9ypmjnblkango0ha
```



```
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/pkbp452l0ulejs73xvfh36jr4

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>docker images
                                                                                                                                   i Info →  U  In Use
IMAGE                                                                                                ID             DISK USAGE   CONTENT SIZE   EXTRA
cloud-negotiation-agent:v1                                                                           870eb6e7dec9        428MB          101MB    U
cloud-negotiation-agent:v2                                                                           870eb6e7dec9        428MB          101MB    U
docker/desktop-cloud-provider-kind:v0.6.0                                                            46d51b990e97        613MB          168MB    U
docker/desktop-containerd-registry-mirror:v0.0.4                                                     1de142695bb0       52.9MB         16.9MB    U
envoyproxy/envoy:v1.36.7                                                                             29d778ba078e        253MB         63.9MB
flaskapp-flaskapp:latest                                                                             da1c91807104        227MB         55.7MB    U
flaskapp-pipeline:latest                                                                             574a7d410748        213MB         52.1MB
flaskapp:latest                                                                                      2aec79eb43b3        213MB         52.1MB    U
gcr.io/k8s-minikube/kicbase:v0.0.50                                                                  b97074569ae9       1.94GB          545MB
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93  eb4fec00e8ad       1.94GB          545MB    U
grafana/grafana:latest                                                                               b28bae15e219       1.94GB          482MB    U
jenkins/jenkins:lts-jdk17                                                                            cce1eeb79902        785MB          280MB    U
kindest/node:v1.36.1                                                                                 3489c7674813       1.31GB          383MB    U
my-flask-app:latest                                                                                  7510e795b10d        222MB         54.3MB
postgres:15                                                                                          74e110c41804        633MB          164MB    U
prom/prometheus:latest                                                                               efd719c99d83        381MB          114MB    U
yashpatelldocker/flaskapp:latest                                                                     da1c91807104        227MB         55.7MB    U

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>
```



```
C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>docker run -d -p 5000:5000 --name my-flask-app-container my-flask-app
9cfdd009c213e2a25dcd7dbfcee6b46bd005689639d617c1f4c3caeb8729b60f

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>docker ps
CONTAINER ID   IMAGE                        COMMAND                  CREATED                        STATUS             PORTS
                                                  NAMES
9cfdd009c213   my-flask-app                 "python app.py"          13 seconds ago                 Up 12 seconds      0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
                                                  my-flask-app-container
4574c5c395ed   jenkins/jenkins:lts-jdk17    "/usr/bin/tini -- /u…"   About an hour ago              Up About an hour   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp,
0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   myjenkins
6469f5fb413a   postgres:15                  "docker-entrypoint.s…"   2 months ago                   Up About an hour   0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
                                                  flaskapp-db-1

C:\Desktop\SYMBI\SEM 7\DevOps-Lab\Projects\Project 8\my-flask-app>
```



Hello, Docker!

Subject Faculty Name: Prof. Arijit Dutta